

2. Designing Experiments for UbiComp

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Outline

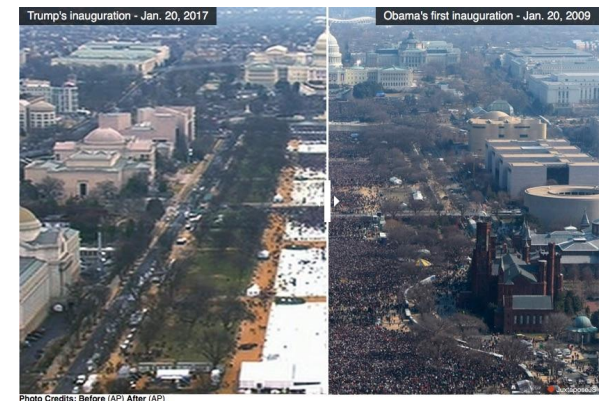
- **Experiment goals**
- Variables: (In-)Dependent variables
- Hypotheses
- Randomness and summarizing the data
- Comparing Values: Significance
- Observational versus Experimental, Quasi-experimental:
 - Designs: One group post-test, pre-test and post-test, interrupted time series, and static group comparison
- Within-subject / Between-group designs
 - Randomization
 - Between-group post-test, pre-test and post-test, Solomon 4-group
- What makes UbiComp different / difficult?
 - Newer evaluation methods

The purpose of studies

- To learn more
- To ensure quality in products
- To compare solutions
- To provide quantitative figures
- To get a scientific statement (instead of personal opinion)



[John Snow & Mapping Cholera](#)



Examples of scientific statements

- 95% of the users can complete the task in less than 3 minutes
- Users did the task quicker with system A, compared to system B
- Users make 10% less errors when using system X, compared to using system Y

Central to Ubiquitous Computing research is the building and analysis of novel computer-based systems; Experiments with users are almost always needed for this.

Why do scientists measure things?

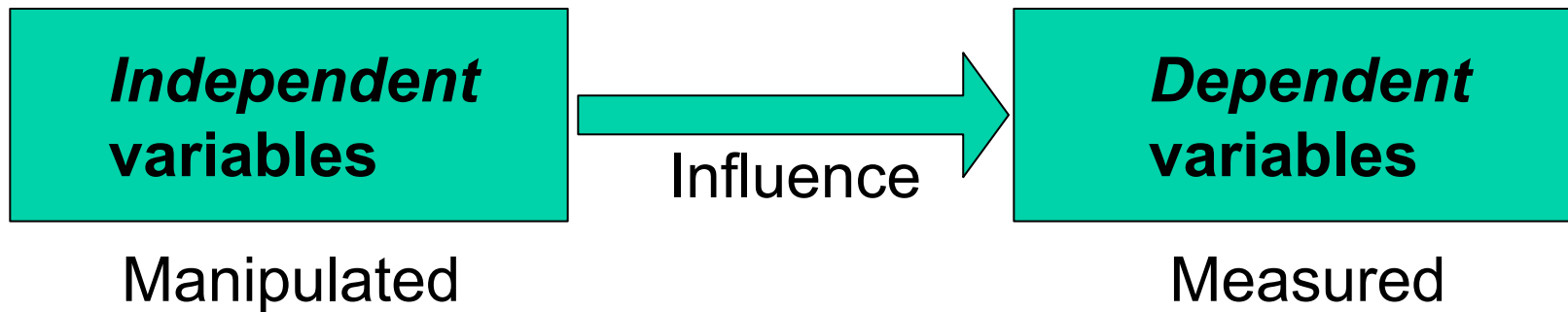
- Find causal links between variables,
e.g. smoking -> cancer



Why do scientists measure things?

- Find causal links between variables,
e.g. smoking -> cancer
- Criteria to **infer** cause and effect (John Stuart Mill, 1843):
 - Cause has to **precede** effect
 - Cause and effect should **correlate**
All other explanations of cause-effect relationship must be ruled out
- Infer causality with two controlled situations:
 - Cause is present (**experimental condition**)
 - Cause is absent (**control condition**)
- Otherwise the situations have to be identical

Defining variables



Defining variables

- **Independent Variables**

- Manipulated by the experimenter
- **Conditions** under which the tasks are performed
- The number of different values used is called **level**
Examples: Users use either speech or keyboard interface (2 levels),
Software is tested with blue, green, or white backgrounds (3 levels), ...

- **Dependent Variables**

- Measured in the user study
- Objective values: e.g. time to complete a task, number of errors
- Subjective values: ease of use, preferred option
- They should only depend on independent variables (conditions)

Hypothesis

- Prediction of the result, before the experiment takes place
- States how a change in the independent variables will influence the measured dependent variables

example: The group of users that use autocompletion in their text editor will produce less grammatical errors, compared to the group of users that do not have this feature

- The experiment will either prove or disprove the hypothesis

Null Hypothesis

- The **Null Hypothesis** predicts that the independent variables do not have any effect on the dependent variables.
If there is a measurable effect, it must be up to randomness.
- The goal of the study is to disprove the null hypothesis (which is easier than to prove the hypothesis):
“If the null hypothesis were true, then the observed effect is very unlikely*. Therefore, the hypothesis is likely true.”
- **Example:** The group of users that use autocompletion in their text editor produced on average 20% less grammatical errors, compared to the group of users that do not have this feature. **This difference is significant (not up to randomness).**

Example: Marshmallow experiment ([Mischel & Ebbsen, Stanford, 1970](#))

Study of delayed gratification in children:

Hypothesis: distracting children increases the time they can wait.

- 50 children (25 boys, 25 girls), age 3.5 - 5.6 (median 4.5)
6 children disqualified for failing to understand task
- 2 experimenters (1 male / 1 female), each conducting sessions with 3 (female / male) and 2 (male / female) children
- 5 conditions:
 - (1) waiting for delayed reward with distracting toy,
 - (2) waiting for delayed reward with distracting ideation,
 - (3) waiting for a delayed reward (no distractor),
 - (4) no waiting for delayed reward, with distracting toy,
 - (5) no waiting for delayed reward, with distracting ideation
- Main result: most important are mechanisms that reduce frustration

Example: Marshmallow experiment ([Mischel & Ebbesen, Stanford, 1970](#))

MEAN MINUTES AND STANDARD DEVIATIONS OF
WAITING TIME FOR A DELAYED REWARD
AS A FUNCTION OF ATTENTION

Statistic	Available for attention			
	No rewards	Both rewards	Delayed reward	Immediate reward
<i>M</i>	11.29	1.03	4.87	5.72
<i>SD</i>	6.84	2.39	6.57	6.43

NUMBER OF CHILDREN WAITING THE MAXIMUM
TIME (15 MINUTES) IN EACH
ATTENTION CONDITION

Situation	Rewards available for attention			
	None	Both	Delayed	Immediate
Not waiting	2	8	6	6
Waiting	6	0	2	2

Qualitative / Quantitative Studies

Qualitative:

- Get “non-measurable” feedback
- General insight
- Used to find:
 - problem areas
 - conceptual errors
 - missing functionality

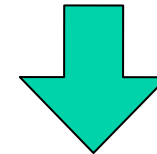


Formative Evaluation:

- Throughout the design
- Helps to shape a product

Quantitative:

- Measure performance
- Generate statistical data
- Used to:
 - verify performance benefits of new devices or interactions
 - determine differences between user groups



Summative Evaluation:

- Quality assurance of the finished product

most useful for

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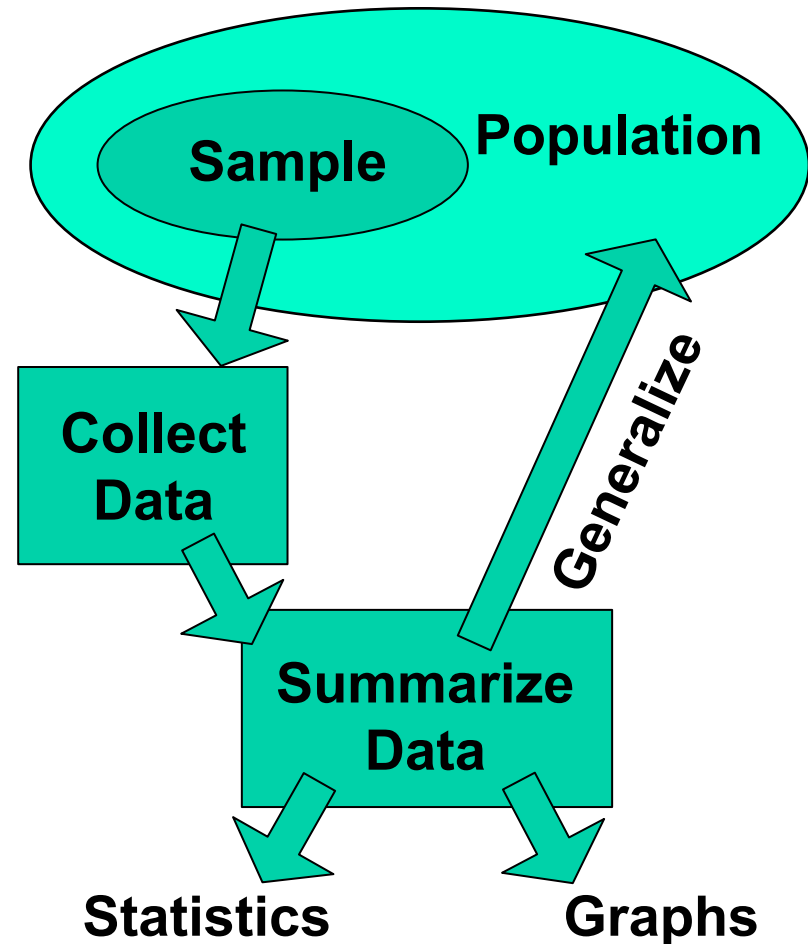
Summarizing data

Collected data needs to be summarized

- to recognize patterns
- to aggregate data

Two ways:

- with **statistics**
- with **graphs**



Unexpected wildness of random — a story*

In the far off land of Gheisra there lies the plain of Nali. For one hundred miles in each direction it spreads, featureless and flat, no vegetation, no habitation; except, at its very centre, a pavement of 25 tiles of stone, each perfectly level with the others and with the surrounding land.

The origins of this pavement are unknown - whether it was set there by some ancient race for its own purposes, or whether it was there from the beginning of the world.

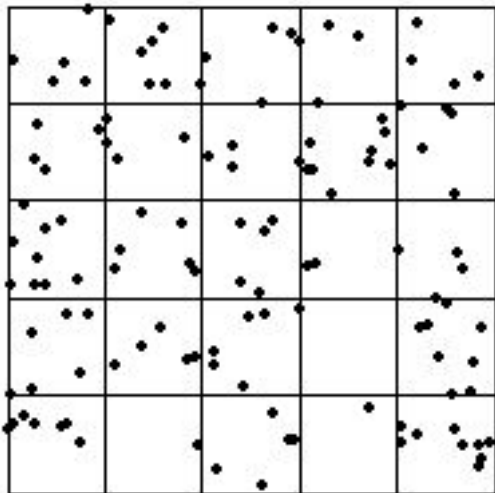
Rain falls but rarely on that barren plain, but when clouds are seen gathering over the plain of Nali, the monks of Gheisra journey on pilgrimage to this shrine of the ancients, to watch for the patterns of the raindrops on the tiles. Oftentimes the rain falls by chance, but sometimes the raindrops form patterns, giving omens of events afar off.

Some of the patterns recorded by the monks are shown below. Which are mere chance and which foretell great omens?

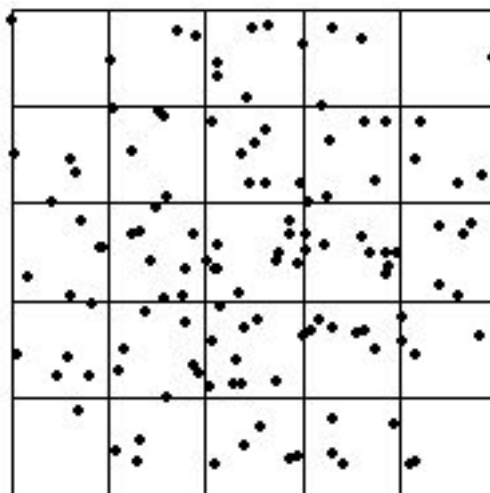
* by Alan Dix

Unexpected wildness of random — a story*

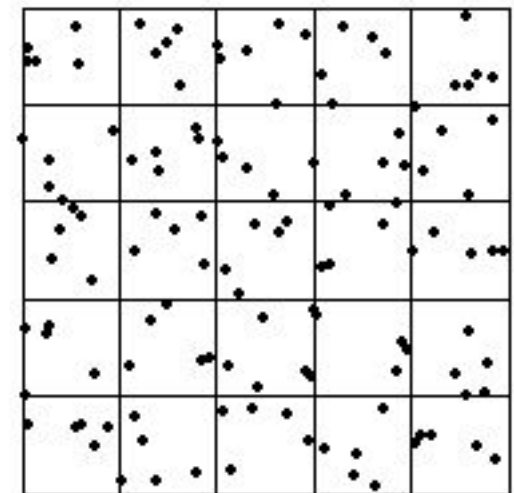
day 1



day 2



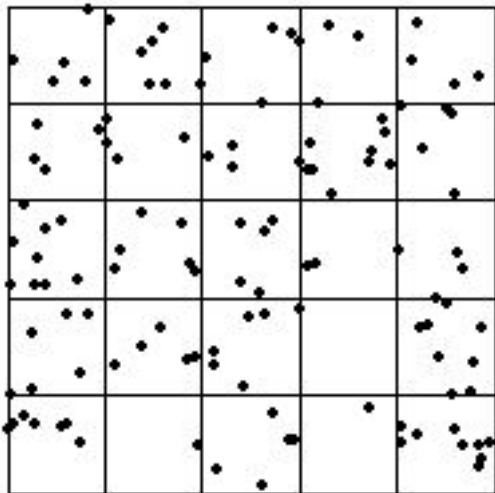
day 3



* by Alan Dix

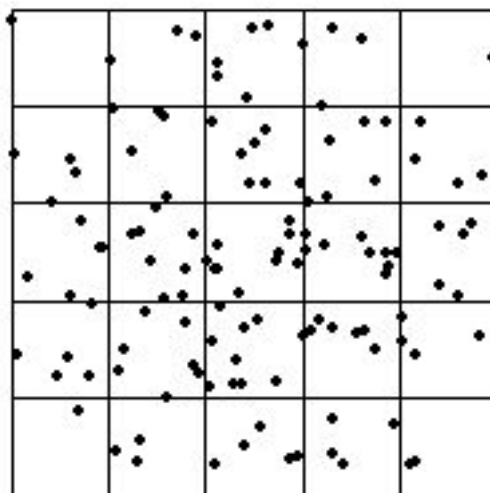
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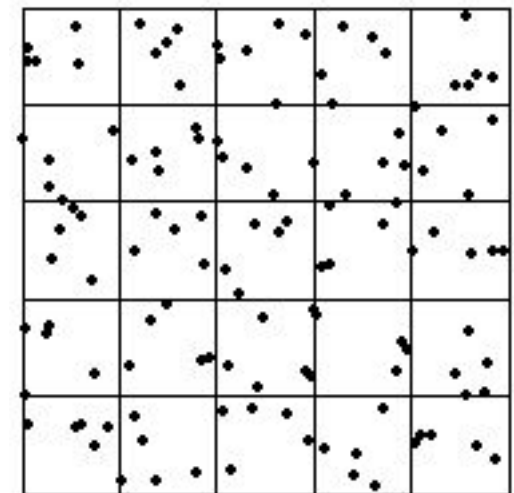
really random, empty
and overfull squares

day 2



random but not uniform,
clumped towards middle

day 3



too uniform, every square
has 5 drops, too good to
be true

* by Alan Dix

Two horse races*

toss 20 coins

add the heads to one row

the tails to another row

the winner is the first row to 10

before we start: what do you think will happen?

* by Alan Dix

Two horse races*

<https://www.meandeviation.com/tutorials/stats/coinrace.html>



Did you get a clear winner?
Or was it neck and neck?

* by Alan Dix

The world is very random

probability head = 0.5

number of heads $\neq \frac{\text{number of tosses}}{2}$

Lessons:

- Apparent differences might be chance
- Real data has some bad values

* by Alan Dix

Finding things out

Problems:

- seeing through randomness
- knowing when you have randomness

Solutions:

- large numbers
- averaging
- measuring variability
- statistical significance

* by Alan Dix

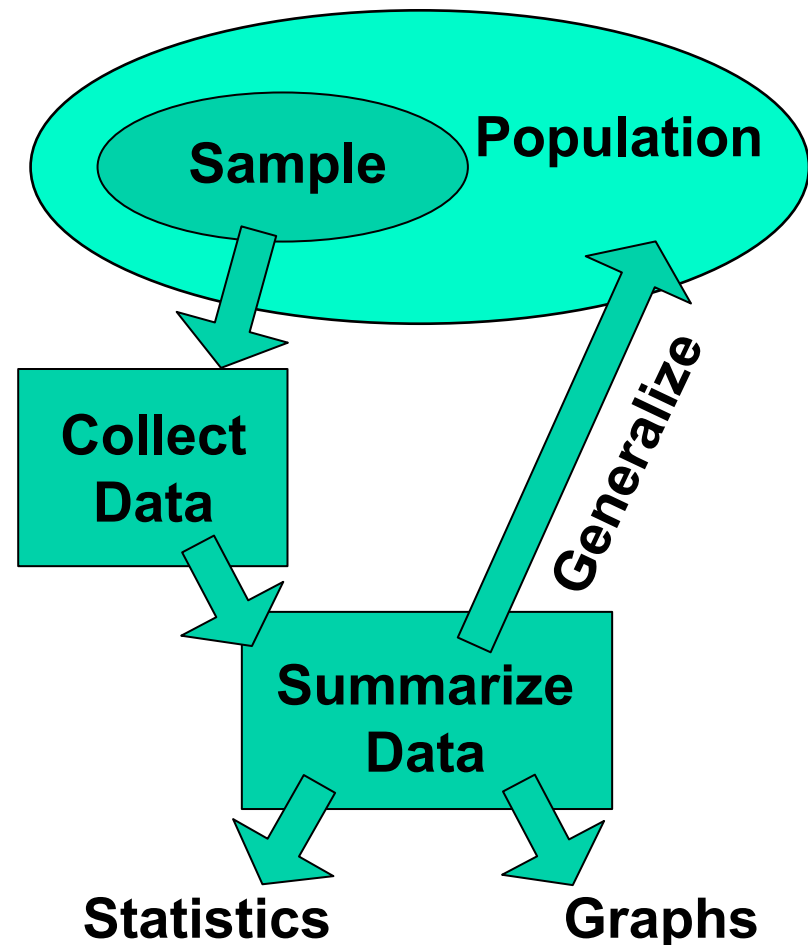
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Summarizing the data

Mode, Median, Mean:

“How can the data be summed up in a single value?”

Example: “Which corporate design do you like most?”

Data:

person 1:	design 2
person 2:	design 3
person 3:	design 3
person 4:	design 5
person 5:	design 2
person 6:	design 3

....

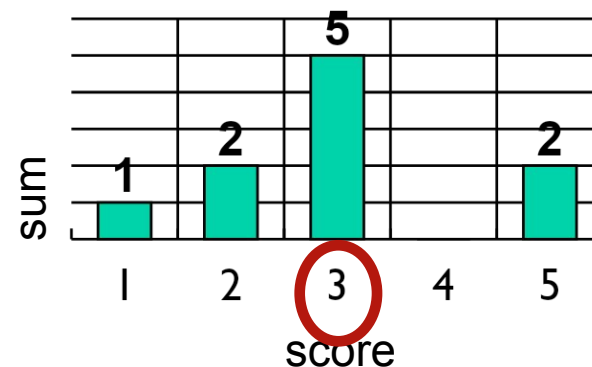
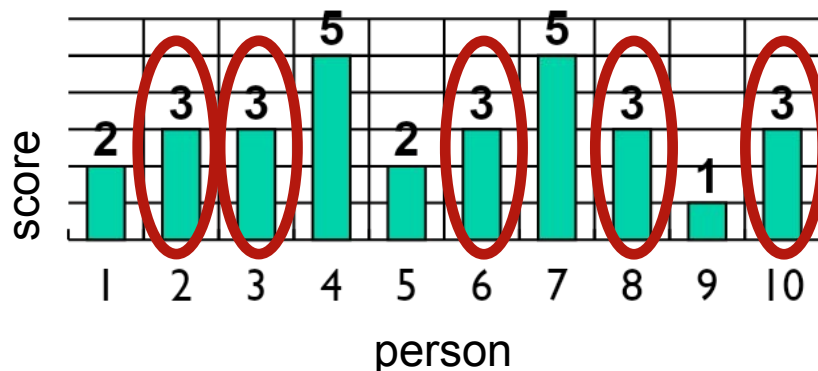
Summarizing the data

Mode, Median, Mean:

“How can the data be summed up in a single value?”

Mode: The most frequent score

- Pros:
 - Easy to calculate and understand
 - Can be used with nominal data
- Cons:
 - There can be more than one mode
 - Mode can change dramatically by adding only one datapoint
 - Independent of all other data in the set



Summarizing the data

Example: “How often did people use our app?”

Data:

over one week:

person 1: 17

person 2: 26

person 3: 53

person 4: 96

person 5: 192

person 6: 23

....

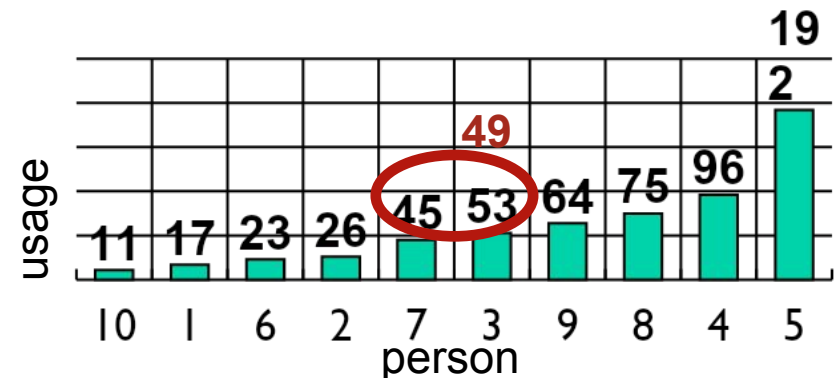
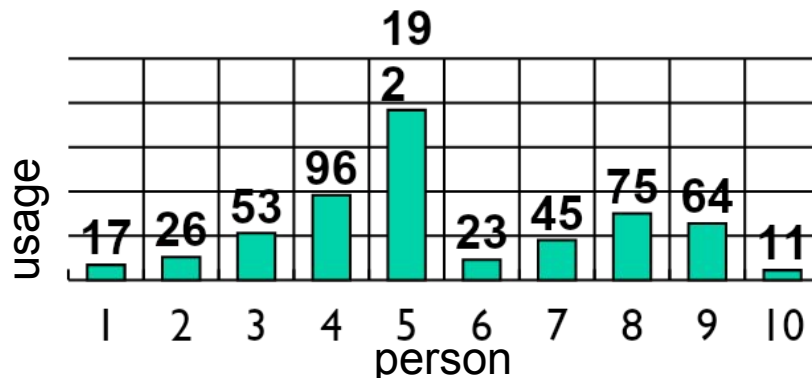
Summarizing the data

Averages: Mode, Median, Mean:

“How can the data be summed up in a single value?”

Median: The middle score*, when sorted by magnitude

- Pros:
 - Relatively unaffected by outliers (very low or high scores) and skewed distributions
 - Can be used with ordinal, interval and ratio data
- Cons:
 - Does not consider all scores of the data set
 - Not very stable
- *: when our data size is even, we take average of two middle scores: $(45+53)/2$



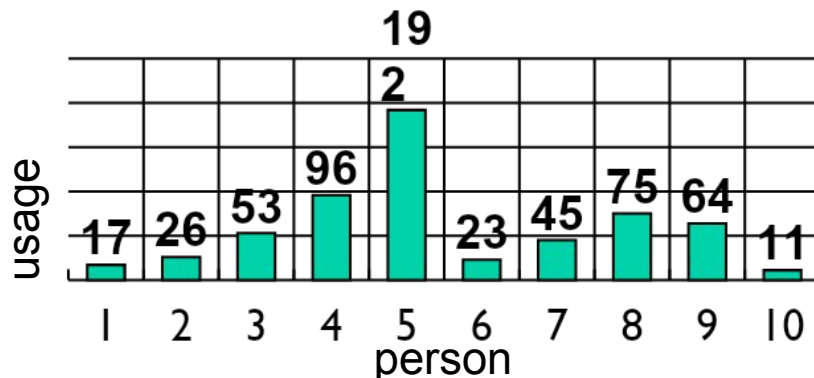
Summarizing the data

Averages: Mode, Median, Mean:

“How can the data be summed up in a single value?”

Mean: The sum of all values, divided by their number

- Pros:
 - Considers every score => most accurate summary of the data
 - Resistant to sampling variation: removing one sample changes the mean far less than mode or median
- Cons:
 - Heavily affected by extreme scores and skewed distributions
 - Can only be used with interval and ratio data



$$\frac{17+26+53+96+192+23+45+75+64+11}{10} = 60.2$$

Summarizing the data

Distributions: Histograms

Example: “time needed to answer all emails in my inbox”

E-Mail: 1, 2, 3, 4, 5, ...

Reply time (minutes): 137.3, 60.1, 120092, 80.9, 3916.06, ...

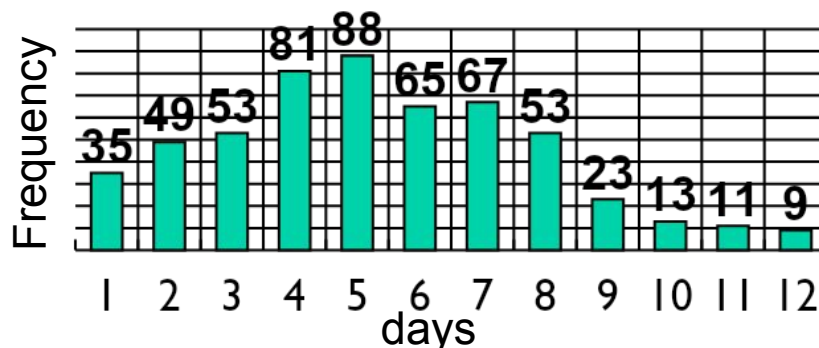
Define a bin size (e.g., days), count the number of occurrences per bin:

0-1 days: 35 emails

1-2 days: 49 emails

2-3 days: 53 emails

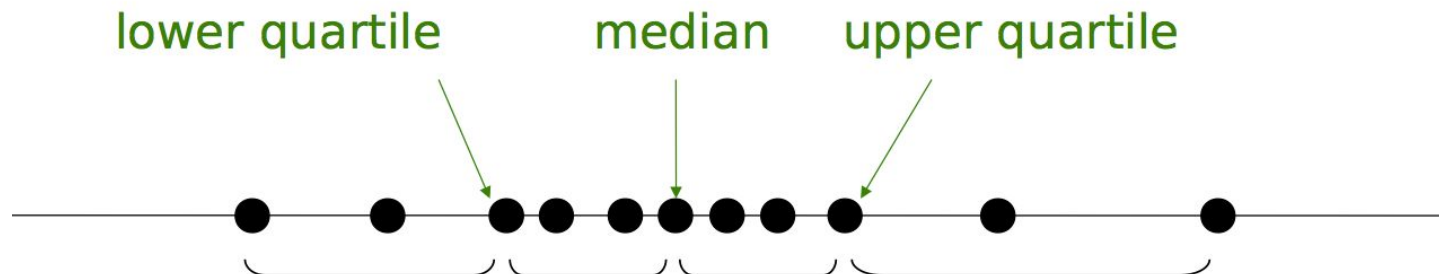
etc.



Summarizing the data

Quantile, Quartile and Percentile

- **Quantile**: 'Cut points' that divide a sample of data into groups containing (as far as possible) equal numbers of observations.
- **Quartile** (Quantile of 4): Values that divide a sample of data into 4 groups containing (as far as possible) equal numbers of observations
- **Percentile** (Quantile of 100): Values that divide a sample of data into 100 groups containing (as far as possible) equal numbers of observations



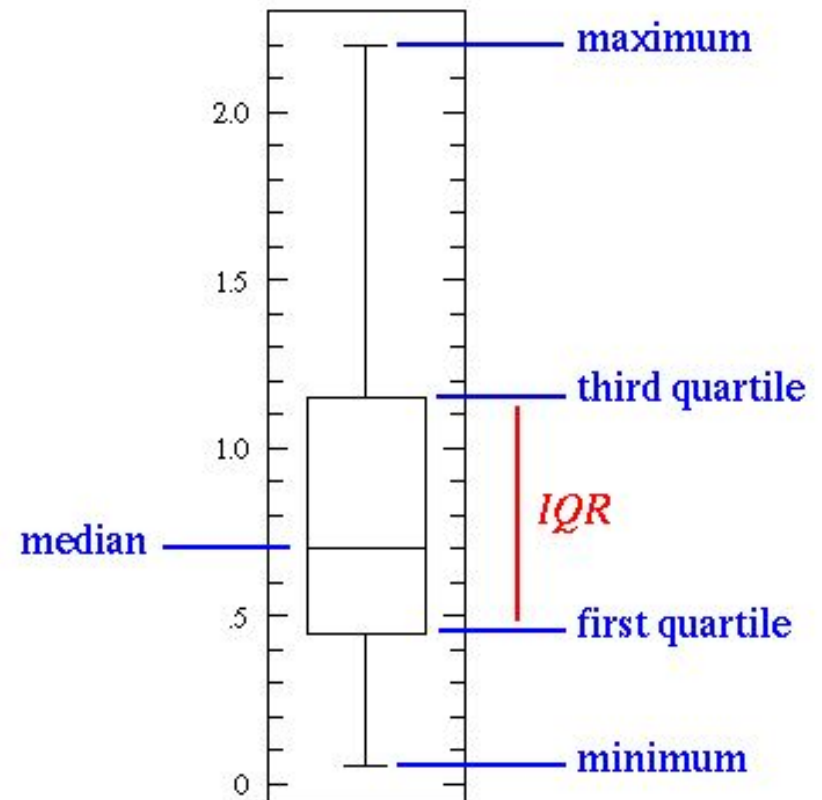
Summarizing the data

Boxplots

or box-and-whisker diagrams

or candlestick charts

show an abstract distribution of data
(e.g., experiment results) in one
visualization

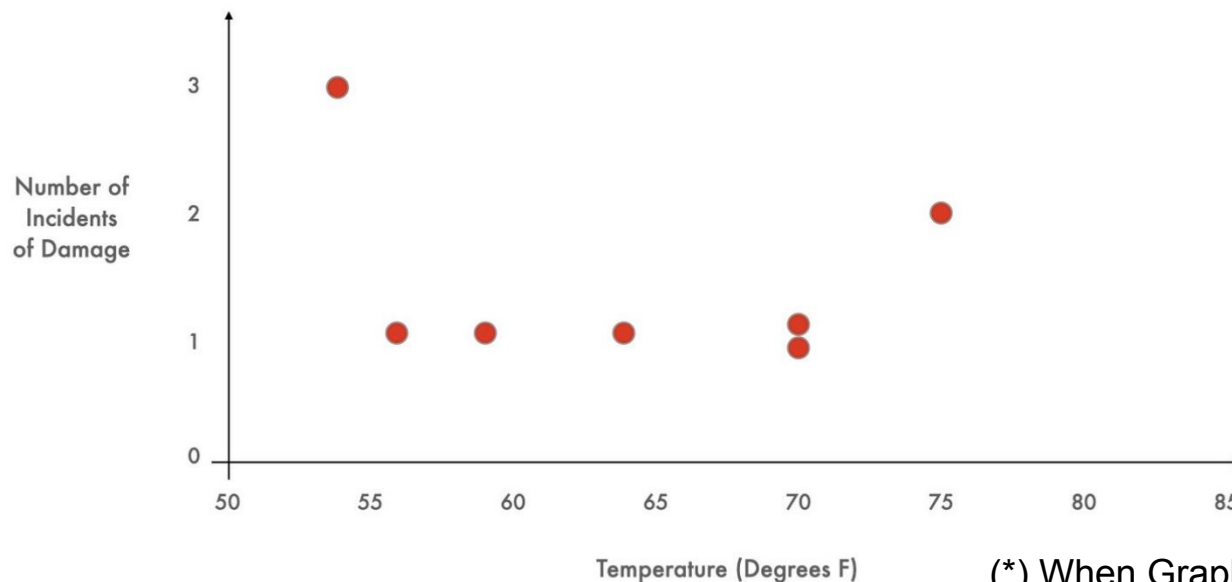


Summarizing the data

Scatter plots

Read the article (*) for the background story

What could be the cause of engine failure?



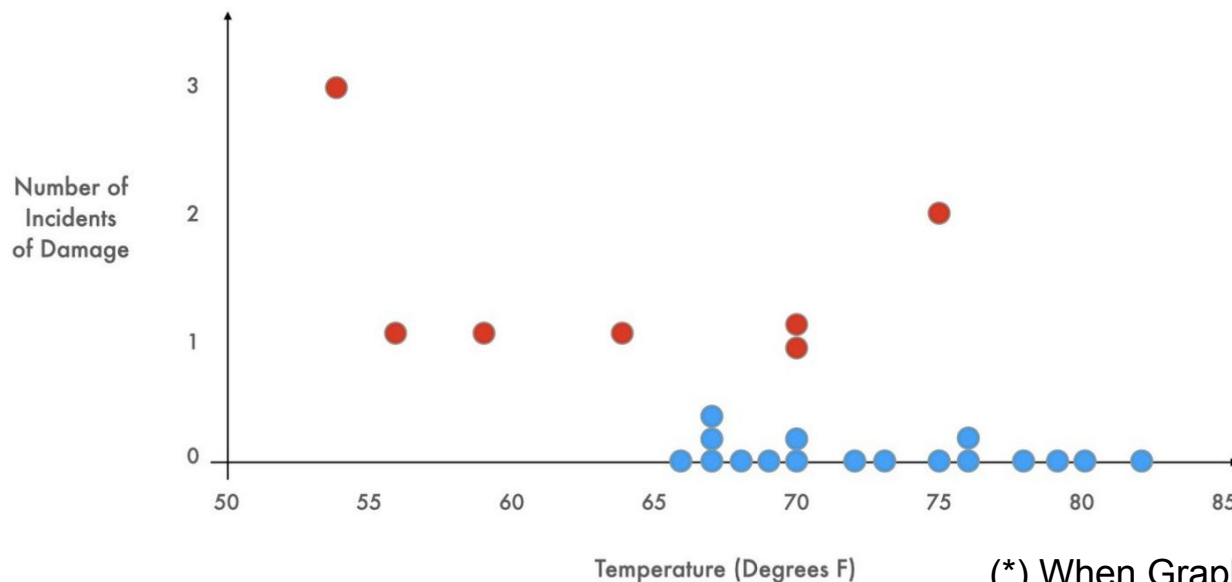
(*) When Graphs Are a Matter of Life and Death, Hannah Fry, 2021 (see moodle)

Summarizing the data

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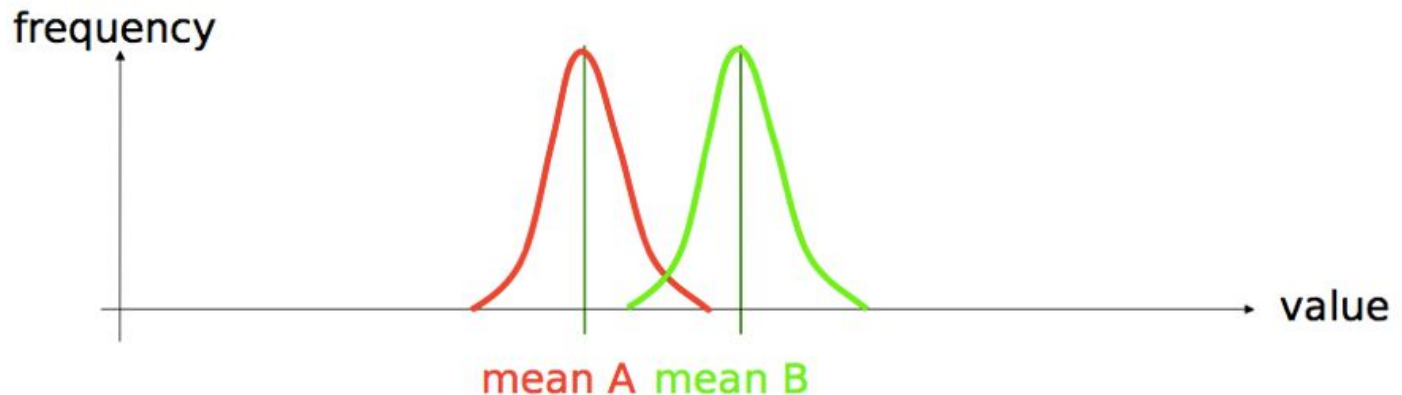
Overall Steps in Designing an Experiment:

1. Consider your **independent** and **dependent variables** and how they are related
2. Write a specific, testable **hypothesis**
→ reformulate as a **null hypothesis**
3. Design experimental treatments to manipulate your independent variable
4. Assign subjects to groups, either **between-subjects** or **within-subjects**
5. Plan how you will **measure** your dependent variable, and what statistical analysis on your data makes sense

Comparing values

Are there significant differences between data from two groups?

2 groups have
small standard
deviations



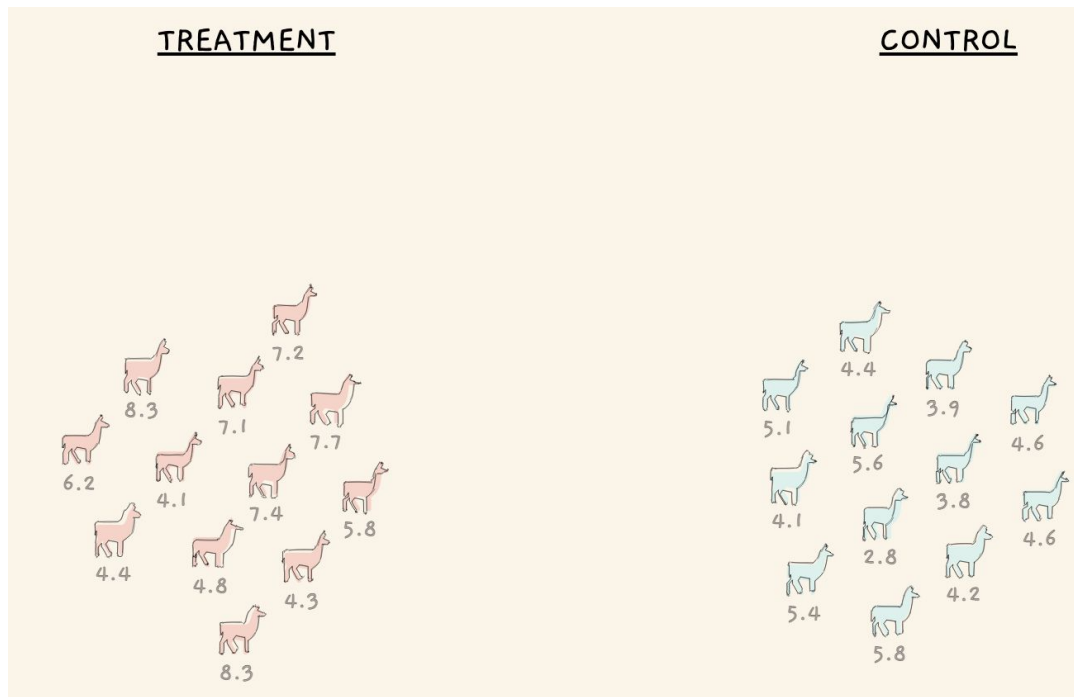
2 groups have
large standard
deviations



Comparing values

Are there significant differences between data from two groups?

Go through the permutation test's explanation on <https://www.jwilber.me/permutationtest/>



Comparing values

Significant differences between measurements?

- In statistics, a result is significant if it is unlikely (probability $p \leq 5\%$) to have occurred by chance
- It does not imply that the result is of practical significance
- Methods such as the **permutation test**, the **t-test** and **ANOVA** (analysis of variance) can be used to calculate p : they give the probability that populations have the same mean (and thus their differences are due to random noise)
- A result of 0.05 from a t-test is a 5% chance for the same mean

Comparing values

Example: The Pepsi Challenge

Participants “blindly” taste cups of Pepsi and Coca Cola, then identify

- **Half the glasses are filled with Pepsi, half with Coca Cola**

Comparing values

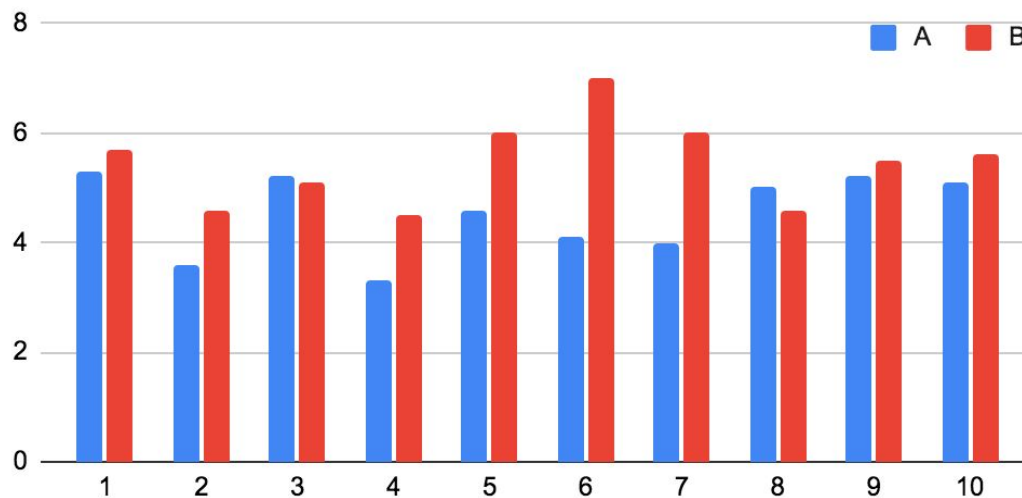
Significant differences between measurements? Example for **significant**

Example #1		
Participant	Method	
	A	B
1	5.3	5.7
2	3.6	4.6
3	5.2	5.1
4	3.3	4.5
5	4.6	6
6	4.1	7
7	4	6
8	5	4.6
9	5.2	5.5
10	5.1	5.6
Mean	4.54	5.46
SD	0.7366591251	0.786271087

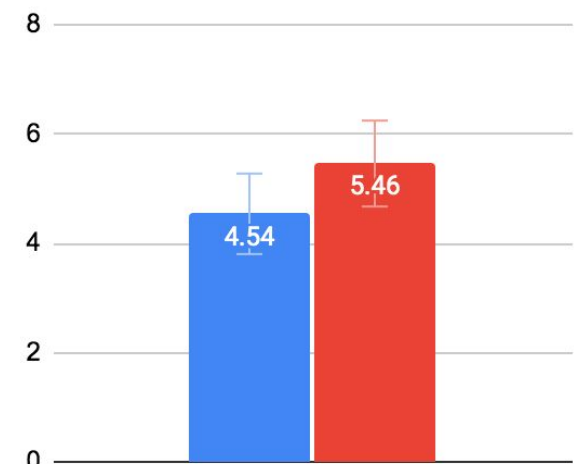
<https://docs.google.com/spreadsheets/d/16gsKpZDFSNrhUw-TVosdi8DHqoAF2VCeN5OF2fscS-4>
 adapted from MacKenzie, Empirical Research in HCI: What? Why? How?

Comparing values

Significant differences between measurements? Example for **significant**



Example #1 / Participant



Example #1 / Mean & SD

TTEST #1:	0.01468355667
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<https://docs.google.com/spreadsheets/d/16gsKpZDFSNrhUw-TVosdi8DHqoAF2VCeN5OF2fscS-4>

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Comparing values

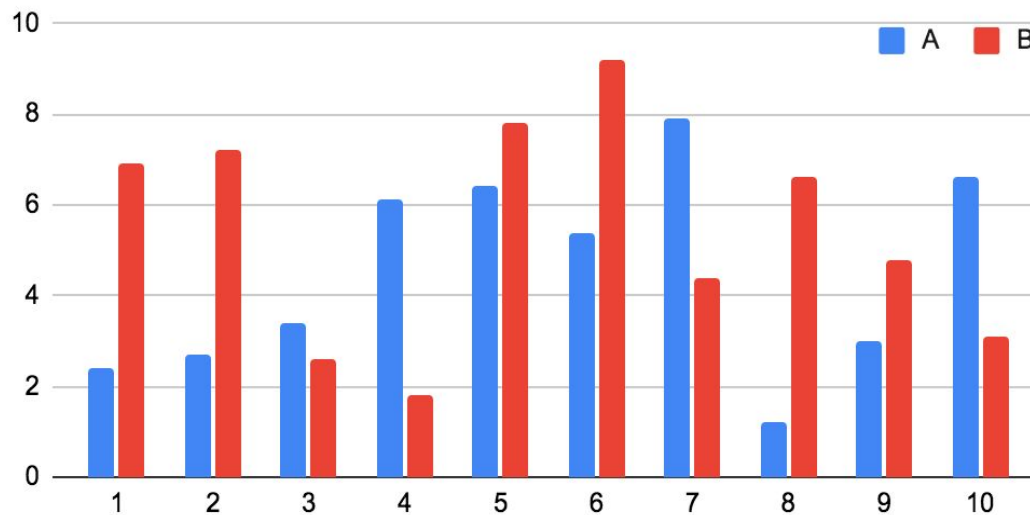
Significant differences between measurements? Example for **not significant**

Example #2		
Participant	Method	
	A	B
1	2.4	6.9
2	2.7	7.2
3	3.4	2.6
4	6.1	1.8
5	6.4	7.8
6	5.4	9.2
7	7.9	4.4
8	1.2	6.6
9	3	4.8
10	6.6	3.1
Mean	4.51	5.44
SD	2.234800513	2.462248295

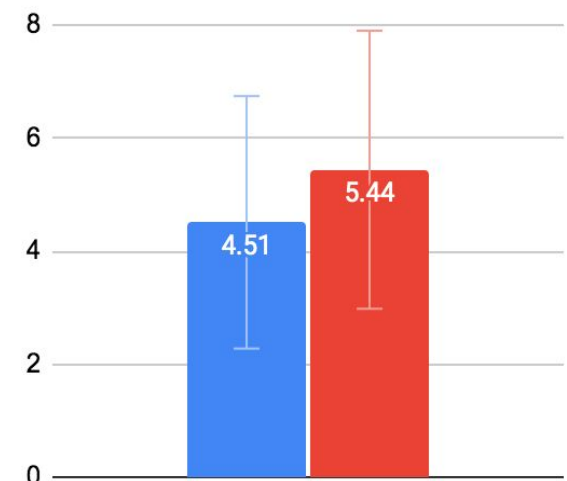
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Comparing values

Significant differences between measurements? Example for **not significant**



Example #2 / Participant



Example #2 / Mean & SD

TTEST #2:	0.3882321006
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Observational vs. Experimental methods

Two approaches to answering research questions:

- **Observational** (= correlational) methods:
Observe what naturally happens in the environment without interfering
- **Experimental** methods:
Manipulate some aspects and observe the effects

	Experimental	Observational
Pros	Isolate and control variables => allow causal statements	Natural setting: observe how people behave normally
Cons	Danger of artificial situations => people might behave differently	• Variables are not isolated • Time consuming

Compromise: first use one, then the other type of method:

- Verify causal hypotheses => confirm findings with more natural observations or
- Identity hypotheses through observations => verify hypotheses in experiments

Quasi-Experimental method

Observational

- No manipulation
- Record behavior systematically and objectively
- Strength: observe people how they behave normally
- Downsides: No identification of cause and effect, time consuming

Quasi-experimental

- Some real experiments are not always possible (e.g., ethical reasons)
- Control over timing of measurement
- No (complete) control over independent variables, *no random assignment of study participants in groups*
=> Impossible to isolate cause and effect

Experiment

- Manipulation by experimenter
- Only way to prove cause and effect

Quasi-Experimental method: Example

Research question:

Does daytime headlight use make motorcyclists more detectable?

Independent variable = daytime headlight, Dependent variable: number of accidents

Experimental design:

Randomly allocate large group of motorcyclists to 2 groups:

- **experimental group**: uses headlight during daytime
- **control group**: does not use headlight during daytime

=> Ethical reasons against this allocation

Quasi-Experimental design:

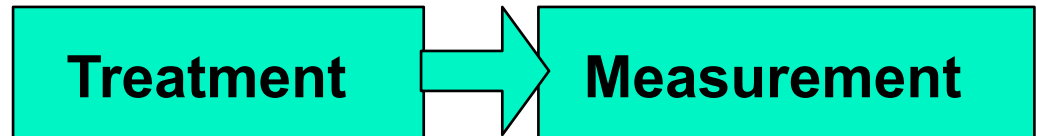
Find motorcyclists with different preferences, then measure.

- Less ethical concerns, but: Pre-existing difference (=> **Group threat**)
Other factors related to the headlights preference can influence the dependent variable (older motorcycles, different safe-conscious levels, ...)

Quasi-Experimental method: Types

1. One group post-test design

- No baseline against which to compare



2. One group pre-test/post-test design

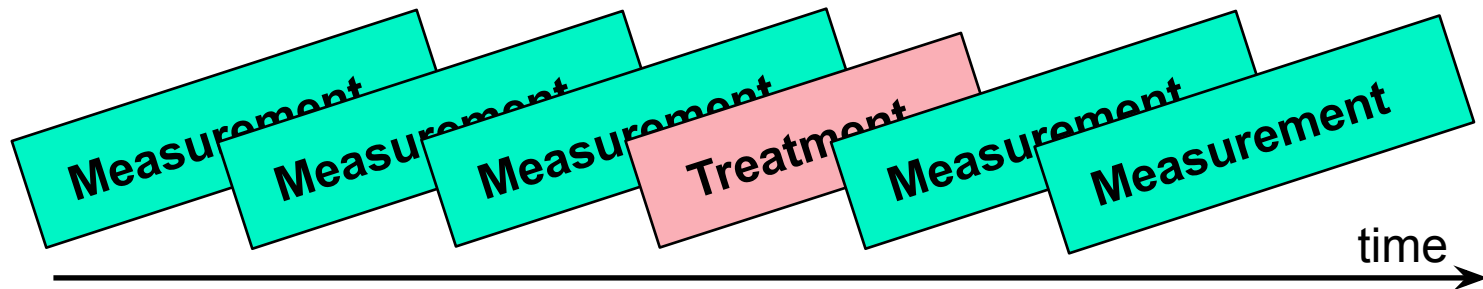
- Assessment of the magnitude of the effect between pre and post test
- No way of telling whether the effect would have occurred without the treatment



Quasi-Experimental method: Types

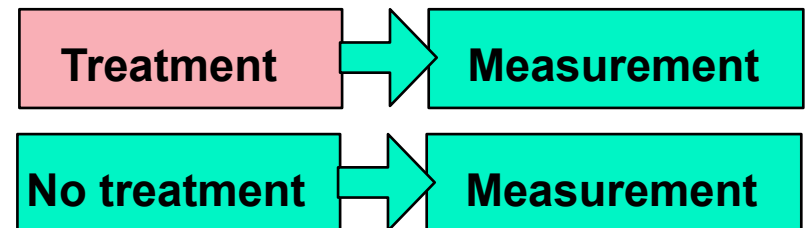
3. Interrupted time-series design

- Series of measurements, some before some after the treatment
- Weakness: Not immune to history threats



4. Static Group Comparison Design

- Experimental group with treatment
- Control group without treatment
- Only weakness: Participants are not randomly assigned (See motorcyclist example)

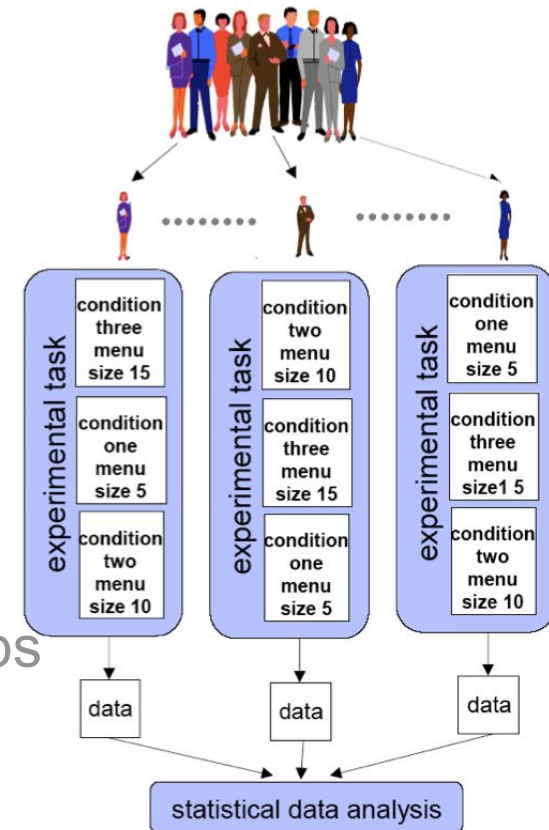


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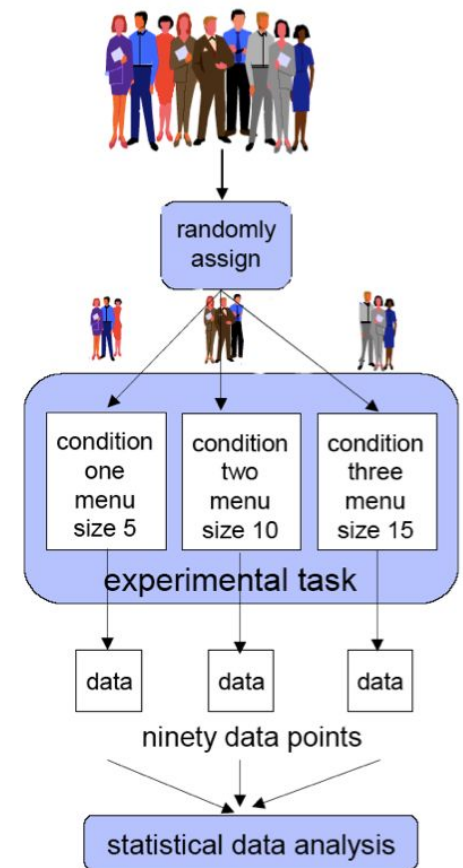
Types of experimental designs

- **within-subject design** => order of conditions
 - („repeated measures“)
 - Each subjects is exposed to all conditions
 - Randomize the order of conditions to avoid ordering effects
- **between-group design** => allocation to groups
 - (“independent measures”)
 - Separate groups of participants for each condition
 - Careful selection of groups is essential



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 - Randomize the order of conditions to avoid ordering effects
- **between-group design** => allocation to groups
 - (“independent measures”)
 - Separate groups of participants for each condition
 - Careful selection of groups is essential



Types of experimental designs

	Pros	Cons
Within subjects	Fewer participants required (n)	• Carry-over (learning) effects • Sometimes impossible (e.g., gender)
Between groups	• No carry-over effects • Less fatigue	• More participants required ($n * [\text{number of conditions}]$) • Usually harder to show significance

Types of experimental designs: Randomization

In all types of experiments *randomization* is crucial:

- In within-subject designs => random order of conditions
- In between-group designs => random allocation to groups

Example for between group design:

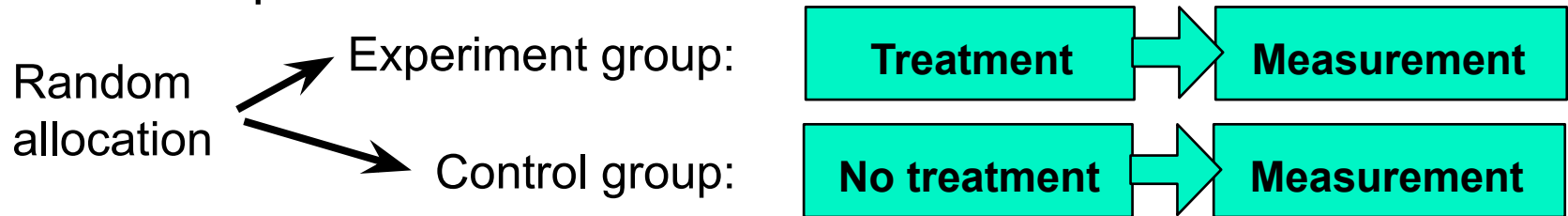
The [Lanarkshire Milk experiment](#) (1930)

- Huge, expensive experiment with 20 000 school children to examine nutritious effects of milk
- Teachers „randomly“ assigned children to:
 - Experimental group: children receive milk every day
 - Control group: children received no milk
- *Teachers tended to assign poor children to the experimental group*
- Result:
 - Control group were by far superior in weight and height
 - Whole study worthless due to the lack of randomization

Types of between-group designs

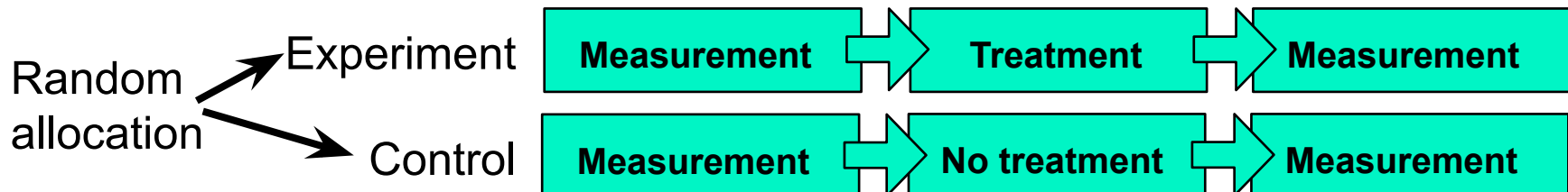
1. Post-test only control group design

- Weakness: no way of knowing whether randomization fails to produce equivalence



2. Pre-test / post-test control group design

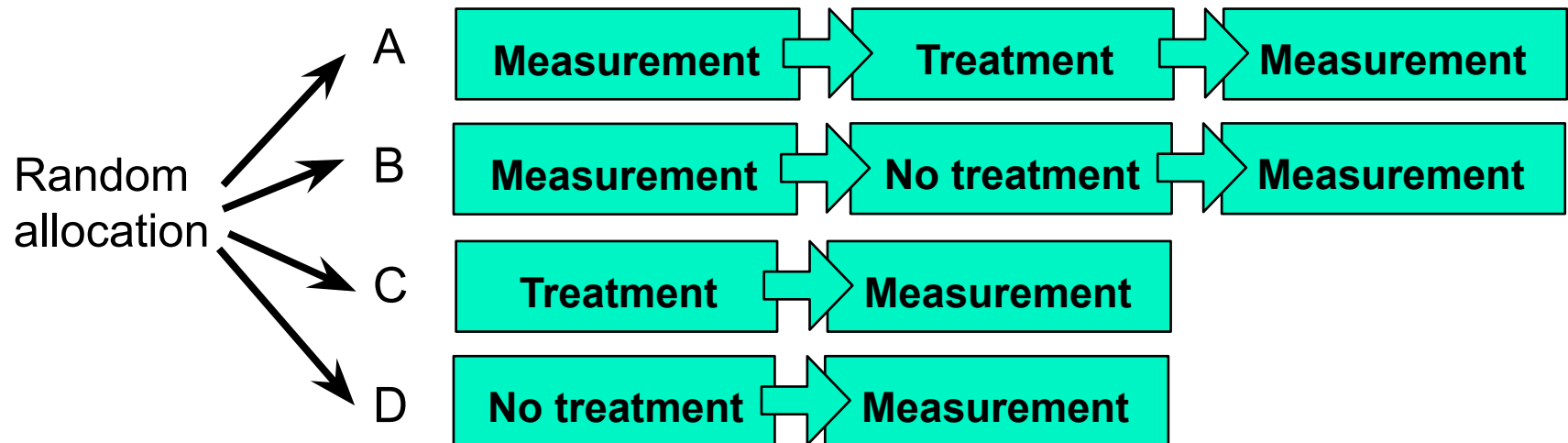
- Pre-test guarantees equivalence
- Weakness: pre-test might affect the performance



Types of between-group designs

3. Solomon four-group design

- Two experimental groups: A and C
- Two control groups: B and D
- Groups C and D show the effects of presence/absence of treatment
- Groups A and B show the effects of the pre-testing
- Very expensive in time and number of participants => rarely used



Outline

- Experiment goals
- Variables: (In-)Dependent variables
- Hypotheses
- Randomness and summarizing the data
- Comparing Values: Significance
- Observational versus Experimental, Quasi-experimental:
 - Designs: One group post-test, pre-test and post-test, interrupted time series, and static group comparison
- Within-subject / Between-group designs
 - Randomization
 - Between-group post-test, pre-test and post-test, Solomon 4-group
- **What makes UbiComp different / difficult?**
 - Newer evaluation methods

Studies in Ubiquitous Computing

- Strong focus on “In the wild” studies [see: Research in the Wild, Yvonne Rogers]:

Human beings have evolved to operate in a complex, dynamic and unpredictable world. In stark contrast, most research about human behaviour has been conducted in controlled lab conditions, focusing on one domain (e.g. cognitive psychology) and within that a specific phenomenon (e.g. spatial memory). The effects of the real-world context are kept to a minimum or eliminated in order to isolate variables and test hypotheses. However, lab experiments account for only 20% of the variance we see in everyday life. This raises serious questions about how ecological valid and generalizable the results are from lab studies and the theories that are derived from them.

- See also the lecture slides on activity recognition for the importance of *real data* as opposed to simulations
- Problem: **A whole system first needs to be built, then thoroughly evaluated**

Wizard of Oz

- Study participants interact with a (computer) system
- Participants believe the system works autonomously
- But it is actually being operated by an unseen experimenter
- “Experimenter-in-the-Loop” technique, researchers can immediately get feedback, without having to develop the whole system
- See also: Mechanical Turk



Mechanical Turk, after Wolfgang von Kempelen (1769)

Living Lab

- User-centred ecosystem where new research and development is routinely integrated / updated
- Parallel research / innovation processes
- Direct experiences and feedback from users, on incremental prototypes



Participatory Design

- All participants cooperate with designers, researchers and developers during research and development of a system
- Roots in **action research** and **sociotechnical design**
- Requires regular interaction between all involved and methods to regulate decisions



Participatory urban planning in Helsinki

Further Reading Papers:

Hypothesis: "greater immigration reduces support for social policies among the public"

Study with 161 social science researchers working on the **exact same data** reach radically different conclusions, almost symmetrically distributed among positive/negative effect.

Of all teams, one conducted preliminary measurement scaling tests, concluded that the hypothesis could not be reliably tested and thus did not design or carry out any further tests.

<https://www.jstor.org/stable/27209022>

On Response Styles in Cross-national Survey:

<https://www.deep-insight.com/wp-content/uploads/2018/09/2006-IJCCM-response-styles.pdf>

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